

Tableau Data Visualization Assignment Report

Itziar Morales Rodríguez

May 18, 2026

Contents

1	Introduction	2
1.1	Methodology	2
1.2	Sources	2
1.3	AI Usage	2
2	Datasets	2
2.1	Dataset 1: State-Level Household Gun Ownership Proxy, 1949–2020	3
2.2	Dataset 2: ATF Gun Traces by State and Firearm Type, 2014–2023	3
2.3	Join Strategy	3
2.4	Analysis	3
2.5	Processing	4
2.6	Research Questions	4
2.6.1	Question 1: Household Gun Prevalence Over Time	4
2.6.2	Question 2: Traced Firearm Type Composition	4
2.6.3	Question 3: Prevalence vs. Traced Patterns	4
3	Visualization Prep	4
3.1	Storytelling	4
3.2	Colors and Themes	5
3.3	Tableau Feature Choices	5
4	Visualizations	5
4.1	Q1 – Household Gun Prevalence by State	5
4.2	Q2 – Firearm Type Composition by State	6
4.3	Q3 – Prevalence vs. Traced Firearm Patterns	6
4.3.1	Visualization Summary	7
5	Conclusions	7
6	References	8

1 Introduction

This project explores the relationship between state firearm legislation and gun violence in the United States, using data spanning 1949-2023. While estimated gun ownership is included as a contextual variable, it is not treated as the primary explanatory factor. Evidence from countries such as Finland and Switzerland, where high gun ownership coexists with low rates of gun violence, suggests that the regulatory environment surrounding firearms may matter more than ownership prevalence alone. This project therefore focuses on whether the strength of state firearm laws is associated with lower rates of firearm homicide and suicide, and whether changes in legislation over time correspond to measurable shifts in gun violence rates.

1.1 Methodology

The project followed the process below to achieve the goal described above:

1. Dataset selection and join strategy design
2. Data analysis and cleaning outside Tableau
3. Research questions derived from data exploration
4. Visualization types needed to answer each question and storytell
5. Data source connection, relationships, and calculated fields in Tableau
6. Tableau feature selection (parameters, filters, actions, tooltips)
7. Dashboard design and interactive storytelling

1.2 Sources

Numerous sources are used throughout the project such as:

- Official dataset documentation and codebooks
- ATF and Harvard Dataverse dataset metadata
- Academic literature on firearm ownership proxies
- Tableau official documentation
- Lecture notebooks and course materials
- AI assistance (declared below)

Regardless of the source, these are indicated through bibliography and references, or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage

AI was used during this project in a critical and supervised manner. All AI-generated suggestions were reviewed and scrutinized before being applied, with particular attention to the risk of hallucination. AI was consulted primarily for dataset pairing strategy, question framing, and Tableau workflow guidance, and was not used to generate data or conclusions directly.

2 Datasets

Two public datasets were selected and joined for this project. Both were required to be different from those used in the Python assignment, and had not been claimed by another student.

2.1 Dataset 1: State-Level Household Gun Ownership Proxy, 1949–2020

The first dataset¹ is a state-year level proxy for household gun ownership in the United States, covering 1949 to 2020. It is hosted on Harvard Dataverse and is derived from the firearm suicide proxy methodology: the proportion of suicides committed with a firearm is used as an indirect estimate of household gun prevalence by state and year.

- **Source:** Harvard Dataverse, Kang et al.²
- **Grain:** One row per U.S. state per year
- **Key columns:** State, Year, Firearm suicide proxy (FSS), total suicides, firearm suicides, population
- **Date span:** 1949–2020 (72 years, 50 states + D.C.)
- **Format:** CSV

This dataset was chosen because it provides a long-term, population-adjusted, and academically validated measure of gun prevalence that can be compared across states and over time. It is not a literal count of guns per household but an indirect proxy widely used in public health and policy research.³

2.2 Dataset 2: ATF Gun Traces by State and Firearm Type, 2014–2023

The second dataset⁴ is published by The Trace via the Gun Violence Data Hub and originates from the U.S. Bureau of Alcohol, Tobacco, Firearms and Explosives (ATF). It aggregates firearm trace requests submitted by law enforcement, broken down by state, year, and firearm type.

- **Source:** The Trace / ATF eTrace system⁵
- **Grain:** One row per state per year per firearm type
- **Key columns:** State, Year, Firearm type (pistol, revolver, rifle, shotgun, other), Trace count
- **Date span:** 2014–2023
- **Format:** CSV (types_by_state.csv)

This dataset was chosen because it provides a direct state-level breakdown of what types of crime-recovered firearms are most commonly traced by law enforcement. It complements the ownership proxy by offering a second lens on firearm patterns: not how many households likely have guns, but what kinds of guns appear in traced-crime contexts and how that varies by state.⁴

2.3 Join Strategy

Both datasets were aggregated and joined on the shared keys state and year. Since the ownership proxy spans 1949–2020 and the ATF trace data spans 2014–2023, the effective overlap window is **2014–2020**. The join was performed in Tableau using a relationship between the two data sources on the matching state-year grain, ensuring the workbook remains reproducible and reviewable.

2.4 Analysis

Prior to connecting the data in Tableau, both files were inspected to understand structure, null patterns, and label consistency. Key observations:

- State names required standardisation across both files (full names vs. abbreviations)
- The ownership proxy file includes several calculated rate columns derived from raw counts and population
- The ATF trace file contains one row per state-year-type combination, requiring aggregation before the join
- The overlapping year range (2014–2020) was confirmed as sufficient for temporal analysis and question answering

2.5 Processing

Key processing steps applied before connecting to Tableau:

- Standardised state name format across both files to enable the join
- Filtered both files to the 2014–2020 overlap window
- Aggregated ATF trace file to one row per state-year with total trace count and per-type share columns
- Retained population-adjusted rates from the ownership proxy for per-capita analysis in Tableau
- Removed U.S. territory rows (Puerto Rico, Guam, etc.) to keep the analysis to the 50 states and D.C.

2.6 Research Questions

Three questions were formulated based on data exploration and storytelling intent:

1. How has estimated household gun prevalence varied by state over time?
2. What firearm types dominate ATF trace summaries by state?
3. Do states with higher estimated gun prevalence show different traced firearm type patterns?

2.6.1 Question 1: Household Gun Prevalence Over Time

Question 1 establishes the baseline of the story. By mapping and trending the gun ownership proxy across states and years, the viewer understands where guns are most prevalent and whether that has changed over time. This provides essential context for interpreting the ATF trace patterns addressed in the following two questions.

2.6.2 Question 2: Traced Firearm Type Composition

Question 2 shifts the lens from prevalence to composition. By exploring which firearm types appear most frequently in ATF trace data by state, the viewer can identify regional patterns in traced crime guns and assess whether certain states are more associated with specific weapon types than others.

2.6.3 Question 3: Prevalence vs. Traced Patterns

Question 3 connects both datasets analytically. It asks whether states with higher estimated household gun prevalence also exhibit different patterns in ATF-traced firearm types. This is framed as an association analysis rather than a causal claim, consistent with the observational nature of both datasets and standard practice in firearm research.³

3 Visualization Prep

Before building charts in Tableau, design and structural decisions were made to align visualizations with the storytelling intent and assignment requirements.

3.1 Storytelling

The project follows a three-act structure. Q1^{2.6.1} opens with the geographic and temporal baseline: where gun prevalence is highest and how it has evolved. Q2^{2.6.2} introduces the second dimension: what kinds of firearms are being traced by state. Q3^{2.6.3} closes with an analytical comparison between the two datasets, asking whether prevalence and type patterns are associated. The viewer moves from observation to exploration to interpretation.

3.2 Colors and Themes

Colors were chosen to encode meaning consistently, directly addressing the feedback from the previous assignment that color was under-used as a visual encoding mechanism:

- A **sequential red palette** was used for ownership proxy maps, where darker shades indicate higher estimated gun prevalence.
- **Categorical colors** were used for firearm types (pistol, revolver, rifle, shotgun), kept consistent across all sheets so the same type always appears in the same color.
- A **diverging palette** distinguishes states above and below the national average in scatter and detail views.
- Backgrounds are light and gridlines are minimal to reduce visual noise and keep focus on data marks.

These decisions were made in line with visualization theory from Unit 1 and with attention to color-blindness accessibility by avoiding red-green contrasts as the sole distinguishing factor.

3.3 Tableau Feature Choices

The following Tableau features were planned and used across this project:

- **Calculated fields:** Gun prevalence quintile, trace rate per 100k population, firearm type share percentage, above/below national average flag
- **Parameters:** Year selector allowing the viewer to update all views simultaneously to a chosen year
- **Filters:** State multiselect filter, year range slider, firearm type selector
- **Dashboard actions:** Click-to-filter from the map to the bar chart and scatter plot
- **Enhanced tooltips:** Display state, year, ownership proxy rate, total trace count, and top firearm type per state
- **Reference lines:** National average line in trend and scatter views

4 Visualizations

A consistent set of calculated fields and a shared color scheme were applied across all visualizations. Field names are self-explanatory (e.g. Ownership Proxy Rate, Trace Rate per 100k, Rifle Share %) to ensure readability and reproducibility.

4.1 Q1 – Household Gun Prevalence by State

Two visualizations answer Question 1^{2.6.1}: a filled map showing geographic distribution of the ownership proxy and a line chart showing how the proxy has evolved over the available time range.

The **filled map** was chosen because the proxy is a state-level measure and geography is the most natural encoding for state-level comparisons. A sequential red palette was applied and a year parameter allows the viewer to explore any year in the 2014–2020 window.

The **line chart** places Year on Columns and the national average proxy on Rows, with individual state lines in light grey and the national average highlighted. A reference line marks the overall mean.

Insights: Both views reveal strong regional patterns. States in the South and Mountain West consistently show higher proxy values, while states in the Northeast show lower values. The national trend shows a gradual decline in the proxy from the mid-1990s to 2020, visible within the 2014–2020 analysis window.

4.2 Q2 – Firearm Type Composition by State

A **stacked bar chart** was used to answer Question 2^{2.6.2}. Each bar represents one state, segmented by firearm type share. Categorical colors encode each type consistently. The stacked bar was preferred over a pie chart or treemap because it allows direct cross-state comparison along a shared axis.

Insights: Pistols dominate trace counts in most states, particularly in the Northeast and Midwest. Rifles show a higher share in Southern and Mountain states. The variation is substantial and provides a meaningful second dimension for the analytical question in Q3^{2.6.3}.

4.3 Q3 – Prevalence vs. Traced Firearm Patterns

A **scatter plot** was used to explore the association between the ownership proxy rate and the rifle trace share across states. Each point represents one state in a given year. Color encodes the gun prevalence quintile, allowing the viewer to identify clustering patterns visually. A trend line and confidence interval were added from the Tableau Analytics pane.

A **dashboard** integrates the Q1 map, Q2 stacked bar, and Q3 scatter with a shared year parameter filter, a state click-to-filter dashboard action, and enhanced tooltips showing the key metrics for each state-year combination. This brings the three questions together in a single interactive view.

Insights: States with higher estimated gun prevalence tend to show a higher share of traced rifles relative to pistols. However, the relationship is not uniform and there is variation within high-prevalence states, suggesting other structural factors are also at play. This is consistent with the observational and proxy-based nature of both datasets, and no causal interpretation is claimed.

4.3.1 Visualization Summary

Question	Chart	Chart Type	Tableau Feature
Q1 ^{2.6.1}	Fig. 1	Filled map	Parameter, sequential color
	Fig. 2	Line chart	Reference line, filter
Q2 ^{2.6.2}	Fig. 3	Stacked bar chart	Calc. field, filter, tooltip
Q3 ^{2.6.3}	Fig. 4	Scatter + trend line	Calc. field, action, tooltip

Table 1: Visualization summary table

Table 1 summarizes:

- 4 chart types: filled map, line chart, stacked bar, scatter with trend line.
- 6 Tableau features used: calculated fields, parameters, filters, dashboard actions, enhanced tooltips, and reference lines.
- 2 data sources joined by state and year in Tableau.

5 Conclusions

Visualizations reveal the following:

- Gun ownership proxy is highest in Southern and Mountain West states and has declined nationally since the mid-1990s [Fig. 1].
- Pistols dominate ATF trace counts in most states, but rifle shares are notably higher in higher-prevalence states [Fig. 3].
- States with higher estimated household gun prevalence tend to show higher rifle trace shares, though this is associative and not causal [Fig. 4].
- The two datasets together provide complementary views: estimated household gun availability and traced crime-gun patterns by state and year.

6 References

- [1] Seokho Kang et al. *State-Level Household Gun Ownership Proxy Dataset, 1949–2020*. Accessed: 2026-05-18. 2023. DOI: [10.7910/DVN/QVYDUD](https://doi.org/10.7910/DVN/QVYDUD). URL: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QVYDUD>.
- [2] Seokho Kang and Taehee Lee. “State-level household gun ownership proxy dataset, 1949–2020”. In: *Data in Brief* 50 (2023), p. 109548. DOI: [10.1016/j.dib.2023.109548](https://doi.org/10.1016/j.dib.2023.109548). URL: <https://www.sciencedirect.com/science/article/pii/S2352340923006480>.
- [3] Seokho Kang et al. “Extending the Firearm Suicide Proxy for Household Gun Ownership”. In: *American Journal of Preventive Medicine* (2023). DOI: [10.1016/j.amepre.2023.06.004](https://doi.org/10.1016/j.amepre.2023.06.004). URL: <https://pubmed.ncbi.nlm.nih.gov/37160211/>.
- [4] The Trace. *ATF Gun Traces (U.S.)* Accessed: 2026-05-18. Source: U.S. Bureau of Alcohol, Tobacco, Firearms and Explosives (ATF) eTrace system. 2025. URL: <https://datahub.thetrace.org/dataset/atf-gun-traces-data/>.
- [5] Bureau of Alcohol, Tobacco, Firearms and Explosives. *ATF Firearms Tracing Guide*. 2017. URL: <https://crimegunintelcenters.org/wp-content/uploads/2017/11/ATF-Publication-3312.13-ATF-Firearms-Tracing-Guide.pdf>.



Figure 1: Q1 Chart – U.S. State Gun Ownership Proxy Map

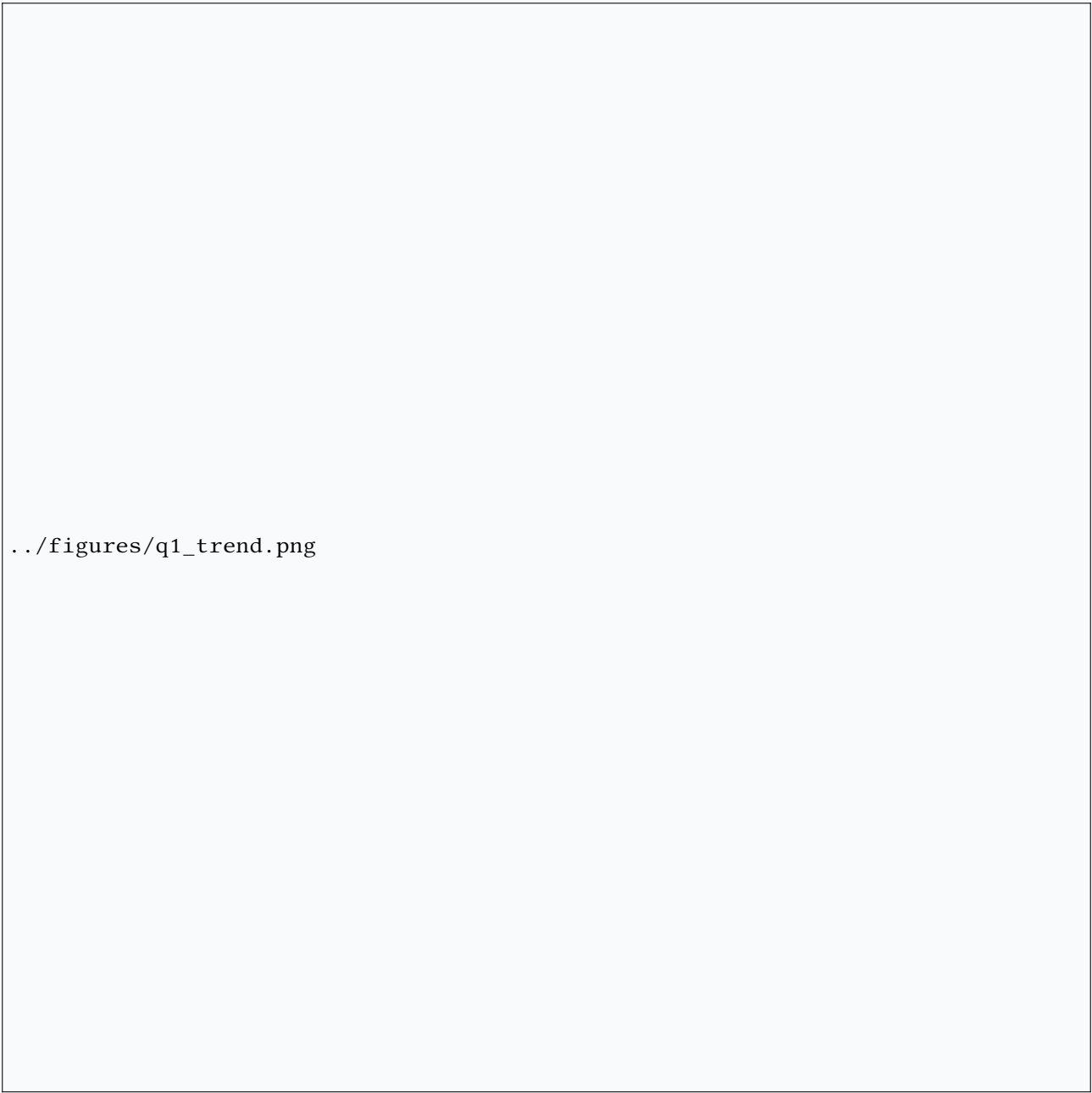


Figure 2: Q1 Chart – Gun Ownership Proxy Trend Over Time



Figure 3: Q2 Chart – Traced Firearm Types by State (Stacked Bar)

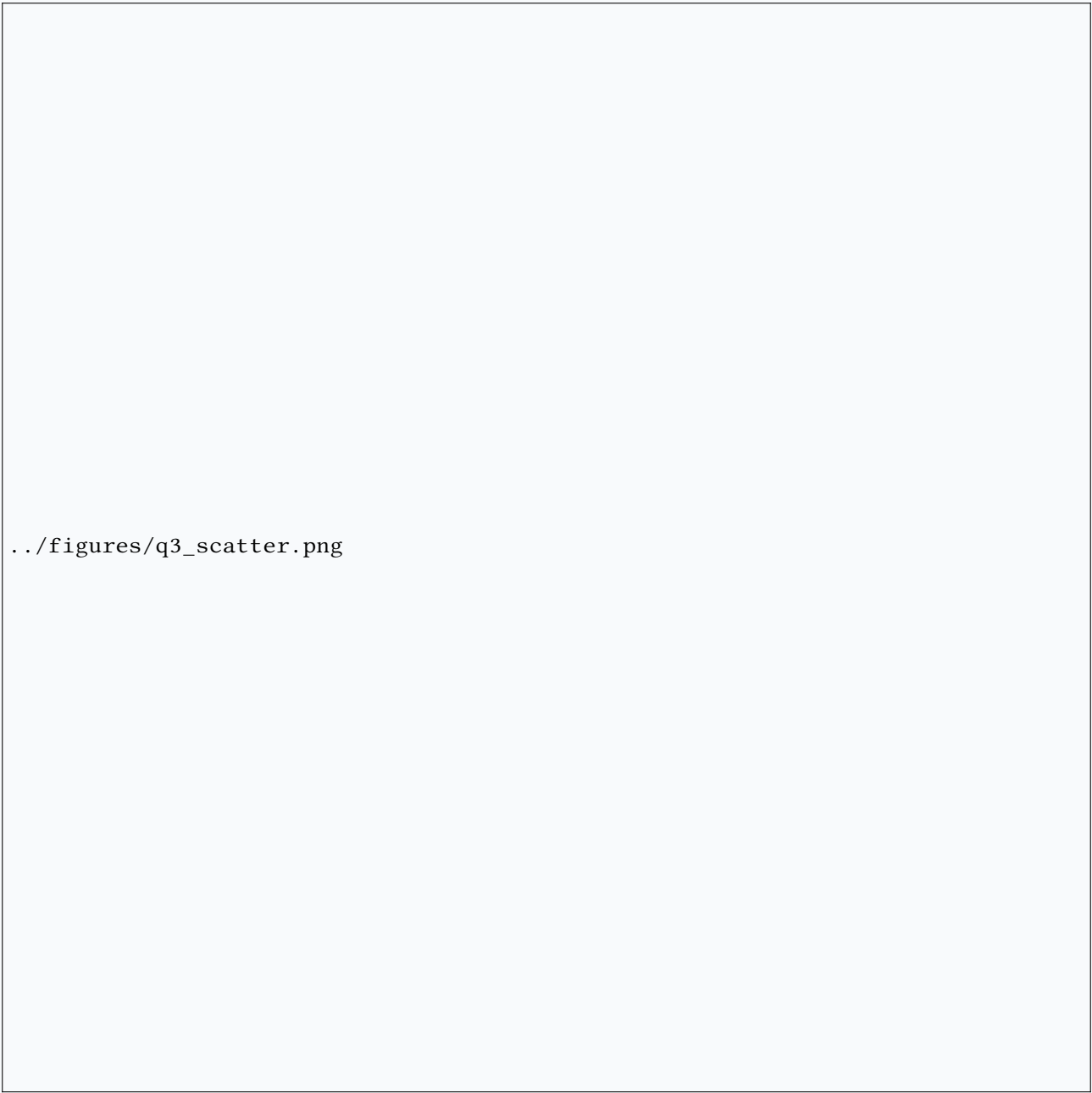


Figure 4: Q3 Chart – Gun Prevalence Proxy vs. Rifle Trace Share